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A fine-tuned large language model based molecular dynamics agent for code generation to obtain material thermodynamic parameters

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In the field of materials science, addressing the complex relationship between the material structure and properties has increasingly involved leveraging the text generation capabilities of AI-generated content (AIGC) models for tasks that include literature mining and data analysis. However, theoretical calculations and code development remain labor-intensive challenges. This paper proposes a novel approach based on text-to-code generation, utilizing large language models to automate the implementation of simulation programs in materials science. The effectiveness of automated code generation and review is validated with thermodynamics simulations based on the LAMMPS software as a foundation. This study introduces Molecular Dynamics Agent (MDAgent), a framework designed to guide large models in automatically generating, executing, and refining simulation code. In addition, a thermodynamic simulation code dataset for LAMMPS was constructed to fine-tune the language model. Expert evaluation scores demonstrate that MDAgent significantly improves the code generation and review capabilities. The proposed approach reduces the average task time by 42.22%, as compared to traditional models, thus highlighting its potential applications in the field of materials science.

Keywords LLM, Agent, Materials science

Calculating thermodynamic properties is a common task in materials computation that provides important insights into material characteristics and informing design and optimization strategies in materials science. For instance, understanding phase transitions in two-dimensional materials is crucial for their applications in electronics and energy storage¹. Accurately predicting and controlling specific thermodynamic properties, including stability and thermal expansion coefficients, is essential for ensuring the material performance in extreme environments, such as those in aerospace applications².

Molecular dynamics (MD) simulations have emerged as an indispensable tool to address these challenges at the atomic scale. Using platforms such as LAMMPS, which is specialized software for MD simulations, researchers can model thermodynamic phenomena, analyze atomic interactions, and investigate material behaviors under varying conditions^{3,4}. MD simulations are particularly effective for studying phase transitions, thermal conductivity, and mechanical properties in both simple and complex systems⁵. However, despite their versatility, MD simulations often involve intricate workflows and demand a high level of expertise and manual input, which can hinder their broader adoption and efficiency^{6,7}. This highlights the need for novel approaches—particularly the integration of AI—to simplify and enhance MD simulations, thus ensuring reliability and precision while improving accessibility⁸.

Recent advancements in large language models (LLMs) have opened new possibilities in this domain. Powerful models, such as ChatGPT^{9,10}, Qwen¹¹, and ChatGLM¹², are becoming increasingly robust. In materials

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science, the integration of AI is gaining momentum, with numerous studies highlighting its transformative potential^{13–18}. For example, ChemLLM¹⁹ fine-tunes LLMs for chemical materials, and MatterGen²⁰ employs generative modeling for inorganic material design. ChemCrow²¹ leverages API tools to tackle chemistry-related problems, and HoneyComb²² explores LLM-based agent systems for materials science, albeit without fine-tuning. Further, AI applications have expanded to specialized domains, such as organic semiconductor device research, with LLM-based AI agents for organic semiconductor device research²³ demonstrating promising results. Similarly, ChatMOF²⁴ uses AI systems to predict and generate metal-organic frameworks. These advancements underscore the growing role of LLMs in advancing materials science research.

However, it is evident from the literature that several issues persist:

- In materials science research¹³, there is a growing demand for high-quality, large-scale data to support advancements in AI-driven research. However, the current data ecosystem is inadequate, with materials data often being scarce, fragmented, non-standardized, difficult to access, and of varying quality, particularly in the area of text-to-code generation. Although some databases and platforms exist, their limited coverage and dispersed data pose significant challenges, thus making it difficult to effectively fine-tune large models for this field.
- Prior studies have mainly focused on text-to-text generation, overlooking the potential of text-to-code generation. As a result, the capabilities of LLMs in theoretical calculations and code development within this domain remain underexplored. This has led to traditional methods where human experts rely on software such as LAMMPS to address domain-specific problems, which demands a high level of expertise, incurring significant time costs. Current text-to-text research fails to efficiently address these challenges.

This paper focuses on using LAMMPS to solve problems related to computing the thermodynamic properties of materials, aiming to establish our proposed approach as a general paradigm that can be extended to text-to-code applications in the field of materials science. To achieve this, this paper introduces Molecular Dynamics Agent (MDAgent), illustrated in Fig. 1, which employs text-to-code generation to automate code generation and correction for LAMMPS, specialized software for MD simulations. Based on a collaboration with domain experts, this paper presents two specialized datasets curated to enhance the capabilities and accuracy of MDAgent. These datasets were for fine-tuning and evaluation in our work.

Our experimental results demonstrate that MDAgent can effectively assist experts in semi-automating domain-specific tasks, serving as a valuable support tool, especially in complex scenarios where it significantly enhances the workflow efficiency. The results also confirmed that this method improves the model's code generation and review capabilities, thus successfully reducing the average time required for materials scientists to complete their tasks.

The main contributions and innovations of our paper are as follows:

- This paper introduces MDAgent, an LLM-based system that uses text-to-code generation to obtain material thermodynamic parameters, reducing task completion time, lowering the knowledge barrier for materials scientists, and providing scalability and cross-task transferability.
- In order to address the lack of publicly available datasets and fine-tuned large models for use in LAMMPS, this paper introduces two specialized datasets: the 'LSCF-Dataset' and the 'LEQS-Dataset'. These datasets are tailored for fine-tuning and testing large models with thermodynamic knowledge in materials science. Our fine-tuning approach enhances the model's review capabilities, making its scoring more accurate and justifications more aligned with expert standards. This also improves the model's ability to generate high-quality, reliable code scripts.

Results

Overview of methodology

MDAgent is an advanced framework designed to automate and enhance the process of obtaining material thermodynamic parameters through MD simulations. The architecture of MDAgent centers around a modular agent system, as shown in Fig. 2a, comprising Manager, Planner, Worker, and Evaluator components, which collaboratively process tasks described in natural language. The Manager interprets user inputs and coordinates task distribution, and the Planner decomposes high-level objectives into actionable subtasks. Workers, specialized for domain-specific software such as LAMMPS, generate corresponding simulation scripts. Evaluators provide feedback on the script quality, iteratively refining outputs to ensure precision and adherence to domain standards.

Two custom datasets were constructed to support this workflow, as illustrated in Fig. 2b: the LSCF-Dataset, focusing on fine-tuning LLMs for accurate LAMMPS script generation, and the LEQS-Dataset, which enables the evaluation of the script quality through expert-annotated scoring and justifications. These datasets address the scarcity of high-quality, domain-specific data in materials science, thus ensuring robust model training and testing.

MDAgent, which combines intelligent automation with human oversight, streamlines complex workflows, reduces the barrier of technical expertise, and accelerates research progress in materials science.

Demonstration of molecular dynamics agent (MDAgent) in thermodynamic property calculations

Four thermodynamic computational tasks were designed to evaluate the effectiveness of MDAgent in preparing LAMMPS input scripts, which are common in material thermodynamics research and have well-established references for evaluation. Input scripts were generated using both MDAgent and an independent method for

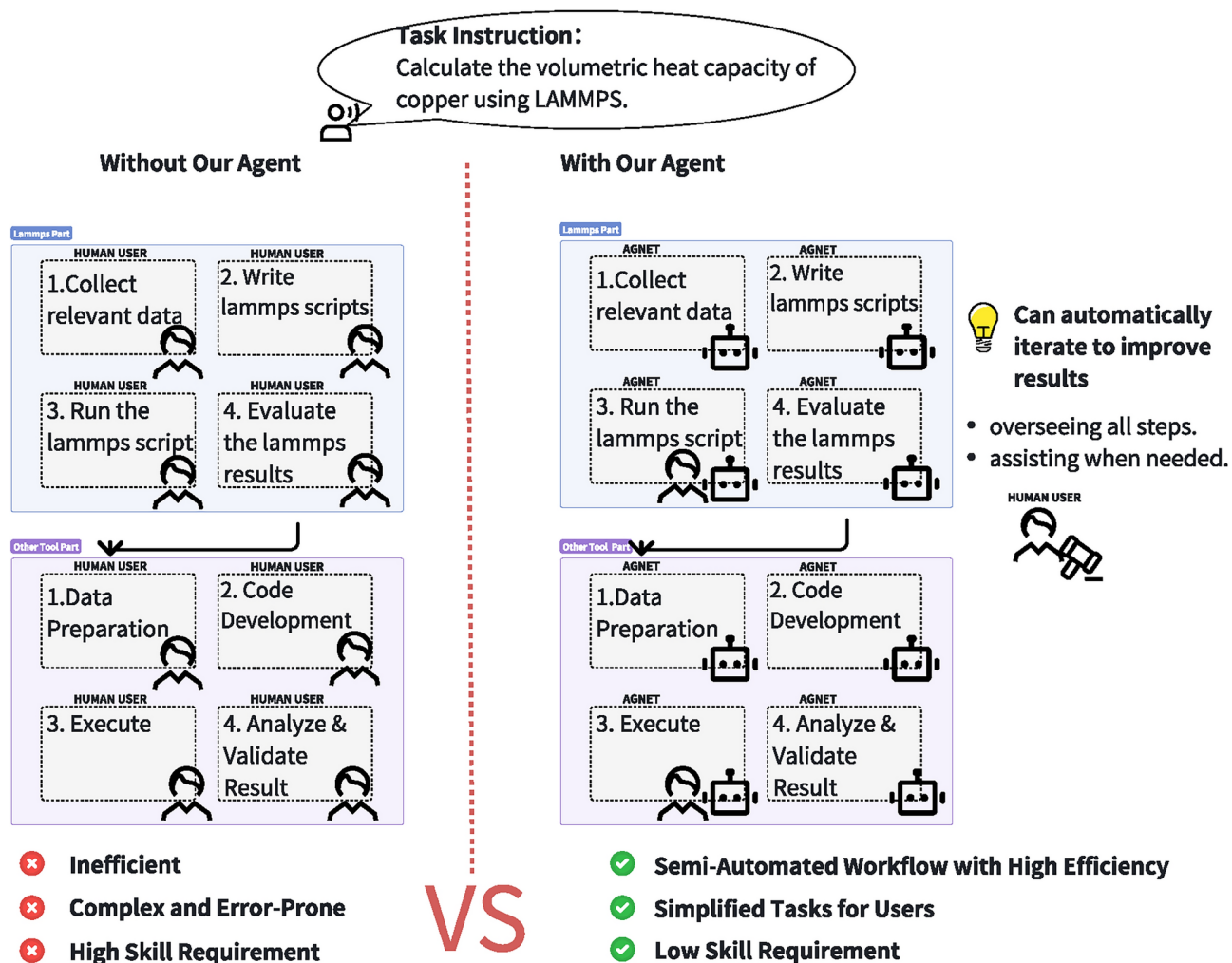


Fig. 1. Comparison of thermodynamic analysis workflow with and without the use of Molecular Dynamics Agent (MDAgent). In the process on the left, the manual workflow requires human users to perform every step, resulting in inefficiency, complexity, and high skill requirements. In contrast, in the MDAgent-assisted workflow on the right, the agent automates key tasks in LAMMPS and other software, enabling a semi-automated, efficient process. The agent simplifies user tasks, reduces the skill requirement, and minimizes errors by overseeing each step and assisting the user as needed.

comparison. The results from the two approaches were analyzed to assess their performances in completing these tasks.

The first task involved calculating the volumetric heat capacity of copper. We extracted the changes in energy and temperature during the simulation, as shown in Fig. 3b, with Fig. 3a depicting the molecular model states at different times during the simulation. The volumetric heat capacity values of copper calculated by MDAgent and the independent method were 3.37 and 3.56 J/(cm³·K), respectively.

The second task was to calculate the equilibrium lattice constant of a diamond crystal. The basic principle was to vary the lattice constant and identify the constant corresponding to the lowest energy, as shown in Fig. 3d. The equilibrium lattice constants obtained with and without MDAgent were 3.52 and 3.54 Å, respectively, with the theoretical value being approximately 3.45 Å. Figure 3c presents the molecular models corresponding to different lattice constants.

The third task involved simulating the melting of copper, where the predicted melting point was identified as the temperature at which a sudden change in the system volume occurred as the temperature increased. The predicted melting points obtained by the two methods were 1440.8 and 1533.5 K, respectively, with the theoretical value being 1357.7 K.

The fourth task was to calculate the linear expansion coefficient of copper. The results calculated by MDAgent and the independent method were 20.6×10^{-6} and 17.5×10^{-6} K⁻¹, respectively, both of which were near the theoretical value.

As shown in Fig. 3, MDAgent demonstrated its capabilities as an intelligent assistant in addressing these thermodynamic property tasks for materials. It effectively supports entry-level materials science practitioners, helping them accomplish tasks more easily and efficiently. However, instead of aiming to outperform human

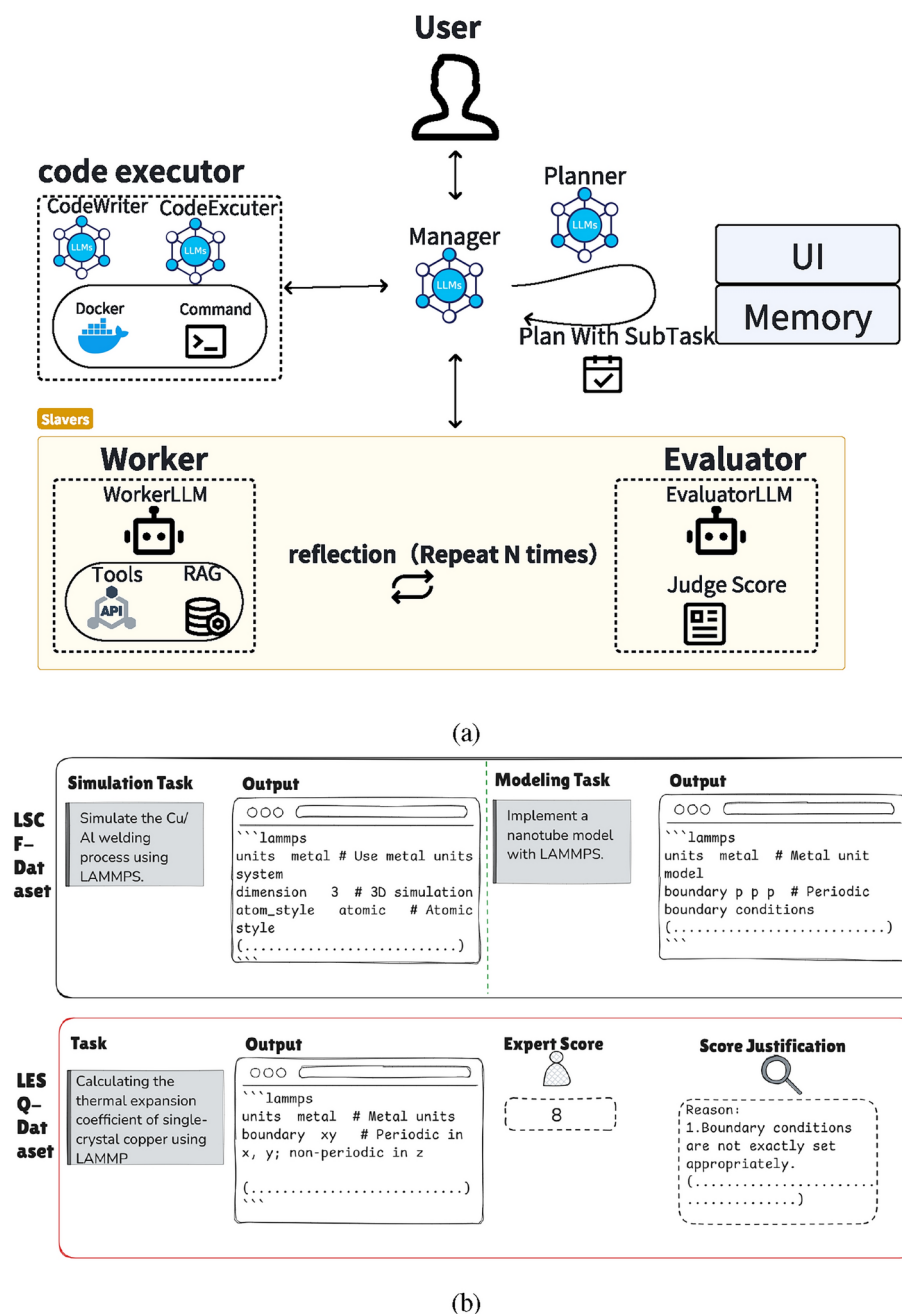


Fig. 2. (a) Architecture diagram: MDAgent with Manager, Worker, and evaluator powered by large language models (LLMs), interacting through a user interface. (b) Example of the dataset used.

experts in the accuracy of calculation results at this stage, MDAgent focuses on enabling novice users to complete tasks quickly while maintaining the required quality level.

Experimental setup

We consulted several experts throughout the experiment who discussed and agreed upon a common evaluation standard prior to proceeding with the assessment, with the aim of minimizing biases commonly found in human evaluations. After the first round of evaluations by all of the experts, a second round of discussions was held to refine and align their opinions, ensuring that the final results reflected a unified consensus. After this, we evaluated the performance of MDAgent using LAMMPS as a case study, focusing on three key aspects:

- (1) **Overall Ability of MDAgent to Assist Users in Completing Tasks:** We conducted user studies with materials science experts to gauge the usability and efficiency of MDAgent. Participants were tasked with defining MD simulations using natural language, and their interactions with the agent were monitored.

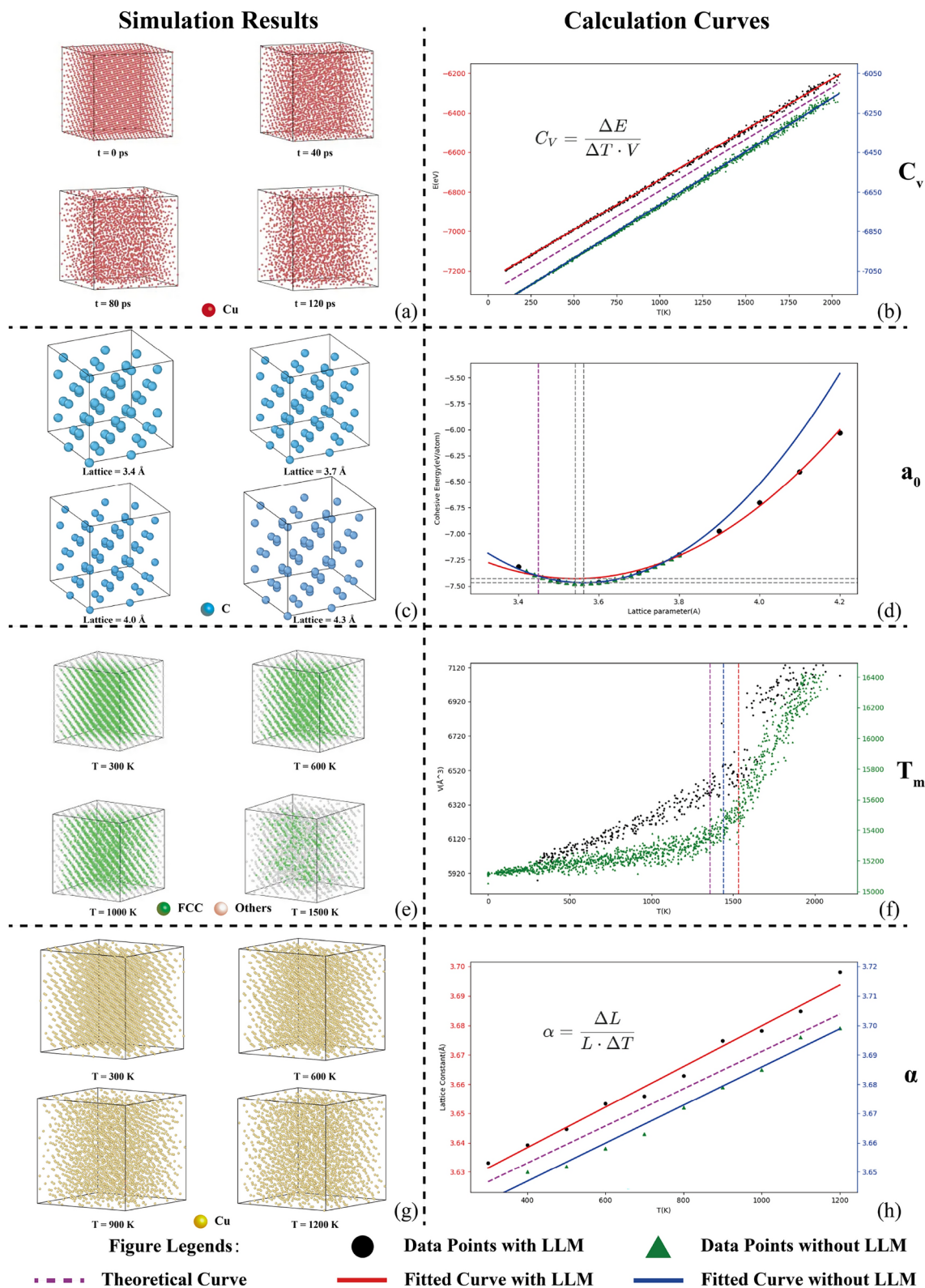


Fig. 3. Simulation models and results: (a) copper molecular model for volumetric heat capacity, (b) energy–temperature curve, (c) diamond crystal model for equilibrium lattice constant, (d) energy–lattice curve, (e) copper molecular model for melting, (f) volume–temperature curve, (g) copper molecular model for thermal expansion, and (h) lattice–temperature curve.

- (2) Ability of LlammpsWorker to Generate Script Code: We evaluated the script files generated by LlammpsWorker for various thermodynamic problems in simple materials. Human experts assessed these scripts for syntactic correctness, adherence to simulation objectives, and alignment with best practices in the field.
- (3) Ability of LlammpsEvaluator to Evaluate Script Code: LlammpsEvaluator was tested on a range of evaluation tasks. It identified issues in existing script code and assigning scores accordingly. The scores that LlammpsEvaluator provided were compared with those from human experts to assess the model's reliability and accuracy in materials science applications.

Evaluation of MDAgent as assistant

Inspired by the experimental design in studies such as ChemCrow²¹, our experiment engaged multiple experts in materials science to complete LAMMPS-related tasks using two different approaches. The first approach used our MDAgent as an assistant, and the second approach relied solely on traditional manual methods based on human expertise. Experts then rated every task based on their satisfaction with every approach, with the total score across all of the methods summing to 10.

The results, shown in Fig. 4a,b, demonstrate that MDAgent is highly effective in assisting experts with the semi-automation of domain-specific tasks. The average task completion time is significantly lower when using the Agent, indicating its ability to help human experts reduce time costs. Further, despite the reduced time cost, human experts still expressed a high level of satisfaction with the Agent's performance.

Evaluation of LlammpsWorker to generate script code

We used a subset of the LEQS-Dataset containing only the task descriptions in order to evaluate the LLMs acting as LlammpsWorker. Task descriptions were input into LAMMPSLLM, which then generated a corresponding output. The results were subsequently reviewed by materials science experts, who assigned scores based on the rating criteria used in the construction of the LEQS-Dataset, providing a score between 1 and 10. The results are shown in Fig. 4c.

Our results demonstrated that the fine-tuned LAMMPSLLM model outperformed other large models of similar size that had not been fine-tuned for specific LAMMPS tasks. Further, the results confirmed the feasibility of using large models to generate LAMMPS scripts for thermodynamic tasks involving simple materials.

Evaluation of LlammpsEvaluator to evaluate script code

We used a subset of the LEQS-Dataset designated as the test set to evaluate the ability of LLMs, acting as the LlammpsEvaluator, to review existing LAMMPS script code and identify errors, effectively simulating the role of expert evaluators.

The scores that LlammpsEvaluator provided were compared with the standard scores given by human experts, and the results were further evaluated by human experts to assess the model's scoring accuracy. The statistical results are presented in Fig. 4d–f. Figure 4d shows that, to a certain extent, human experts acknowledged the evaluation capability displayed by the model. Figure 4e,f demonstrate that fine-tuning reduced the mean absolute error (MAE) and mean squared error (MSE) values, indicating the promising potential of using LLMs as evaluators. While LlammpsEvaluator is not yet ideal in terms of performance metrics because of limitations in the parameter size and data volume, it has demonstrated the potential to act as a reasonable evaluation tool. To a certain extent, in the future, with further fine-tuning, it is expected that LLMs could achieve a high level of consistency with human evaluators' ratings, enabling them to accurately identify and correct issues in a manner similar to humans.

Discussion

In evaluating MDAgent, the following three aspects were analyzed. (1) Task Completion Efficiency: MDAgent showed substantial efficiency gains in both task definition and execution, demonstrating an enhanced workflow that reduced the time and effort compared to traditional methods. (2) LAMMPS Script Generation: MDAgent outperformed general models because of targeted fine-tuning, proving its ability to generate precise scripts and demonstrating excellent capabilities in script code generation. (3) Error Detection in Scripts: The LLM used as an evaluator exhibited consistency in scoring and provided constructive feedback on common errors. Overall, we conclude that MDAgent can effectively assist novice materials scientists in solving related problems based on these experiments.

Looking ahead, there are several promising directions for future research. First, we plan to continue expanding the two datasets presented in this paper. Concurrently, we are exploring the extension of the MDAgent methodology to first-principles calculation tasks, including the use of the VASP software package for Density Functional Theory (DFT) calculations aimed at material property prediction. Second, inspired by the concept of "Agent-as-a-Judge"²⁵, we believe that the Evaluator can further enhance its code review capabilities by adopting similar strategies.

Finally, it is important to acknowledge that MDAgent is currently a semi-automated intelligent assistant, because current local small-parameter LLMs are not yet sufficient to automatically run all complex codes and programs, so human involvement is still needed in the process. We plan to transition it into a fully automated system, with a focus on enhancing Docker and LAMMPS integration and increasing the precision of the LLM-driven tool usage. Research such as AutoGPT²⁶, Openai's Operator and TaskWeaver²⁷, which are designed to enable autonomous AI execution by translating user requests into executable code and coordinating various plugins to accomplish tasks, provide valuable insights for our work.



Fig. 4. Comparison of (a) task elapsed time, (b) subjective ratings by experts on task assistance, (c) evaluation scores for mainstream and fine-tuned models, and (d) scores for the Evaluator LLM. (e) Mean absolute error and (f) and mean squared error.

Methods

Detailed description of the code implementation details please see the GitHub repository.

Dataset for LAMMPS fine-tuning and evaluation

Relevant datasets are required to fine-tune and evaluate MDAgent; however, there are currently no suitable open-source datasets available. In addition, there are no well-established automated evaluation methods. LAMMPS was used as a case study, to address this gap, and two MD datasets were developed, namely, the LEQS-Dataset and the LSCF-Dataset. An example of these datasets is shown in Fig. 2b. Based on existing research^{28–30}, the quantity of data in these two datasets is sufficient to enable effective fine-tuning.

LSCF-dataset

We fine-tuned the model using a dataset of LAMMPS scripts in order to improve the LLM's ability to handle LAMMPS material simulations. Because no standard dataset exists publicly, we created the "LAMMPS Script Collection for Fine-Tuning Dataset (LSCF-Dataset)".

We faced several challenges during dataset construction. First, material simulations cover a wide range of issues, requiring diverse data for the model to generalize across different scenarios. However, no integrated LAMMPS script dataset exists, and available scripts are scarce, varied in quality, and scattered. Second, material simulations involve complex terminology and concepts, the correct understanding of which is crucial for improving LLM performance. Third, high-quality data are essential, but they are difficult to obtain due to varying experimental conditions and research methods. Finally, many scripts lacked annotations, thus hindering the model's ability to properly learn terminology, concepts, and script-writing techniques.

We processed the collected LAMMPS scripts through several steps in order to create a high-quality fine-tuning dataset. We gathered case code from diverse sources, including official documentation, published papers, and open-source projects, which are classic and reliable script codes that add diversity and reliability to the dataset. We then screened the code, removing erroneous code through expert review or validation with experimental results, and retained only executable scripts. Finally, we annotated every script and divided it into three main parts: initialization, modeling, and computation, and explained important steps and concepts to help model learning. Then we converted it to Alpaca format for fine-tuning data. The molecular dynamics processes accomplished by each script are summarized as the value of instruction to the data, and the value of output is complete annotated script, with hint at the input section for more complex cases, such as "use NVT system synthesis".

Following processing, we created a dataset of 167 scripts, all of which were sourced from manual production, LAMMPS documentation cases, and online resources at a 1:2:2 ratio. Their structure is shown in Fig. 2b. It included 127 simulation scripts and 40 modeling scripts, covering a wide range of application scenarios and scientific problems. Incorporating a diverse range of problems from various domains enabled the comprehensive dataset to enhance the LLM's capabilities, improving its performance not only in the obtainment of material thermodynamic parameters, but also extending its potential across other fields.

The simulation dataset involves dynamic processes such as temperature change, shape change, force change, etc. It contains problems in various fields such as material mechanical property simulation, material synthesis and processing simulation, interface simulation, fluid simulation, heat transfer simulation and so on. The modeling dataset contains modeling problems for complex material systems (e.g., high-entropy alloys), extreme geometrical topologies (e.g., diamond trigonal tools, sinusoidal rough interfaces), multi-phase heterogeneous interfaces (Si/Cu pre-positioned cracks, CO_2 -encapsulated water spheres), and functionally oriented structures (e.g., nanoindentation of semi-ellipsoidal indentation heads, graphene desalination slits). The broad coverage of different application scenarios and scientific problems ensures that the dataset reflects common situations in molecular dynamics.

LEQS-dataset

In collaboration with experts in materials science, we have developed a dataset: LAMMPS-Expert Question with Score Dataset (LEQS-Dataset). It consists of a collection of quadruples, where each quadruple includes the following:

- *Task* A description of a task conducted using LAMMPS for calculating a specific thermodynamic property of materials using MD.
- *Result* The corresponding LAMMPS script generated to complete the task.
- *Expert score* A manually assigned score, ranging from 0 to 10, given by experts based on unified evaluation rules.
- *Score justification* A detailed justification from the experts, explaining the score and considering multiple factors.

The LEQS-Dataset was constructed through a multi-stage process. Initially, senior materials scientists designed tasks encompassing core aspects of material thermodynamic property calculation, including thermal expansion coefficients, thermal conductivity, density, and phase transition behavior. These tasks served as input for LammmpsWorkerLLM, which generated corresponding LAMMPS scripts. To preserve the LLM's raw output, these scripts were included in the dataset without modification. Subsequently, a team of experts discussed and developed a structured evaluation rubric with clear and uniform dimensions and scoring criteria, also summarizing common error types observed in LAMMPS scripts. This rubric, based on a perfect score of 10, guided the assessment of potential deductions based on six key factors: syntax correctness, logical consistency, parameter rationality, core logic accuracy, logical completeness, and code completeness.

A random subset of the LEQS-Dataset will be used for fine-tuning the LLM acting as the LammmpsEvaluator, with a separate random subset designated for testing to ensure no overlap between the two.

MDAgent design

In this paper, we focus on LAMMPS as a representative software package to demonstrate the feasibility of our approach. Based on established scaling law theories in existing research^{31–35}, MDAgent possesses inherent scalability, and the methodology can be readily transferred to other tasks in computational materials science. By adapting the datasets used for the Workers and Evaluator, along with the underlying models, MDAgent can be trained to generate code for other specialized software packages in the field, thereby addressing a wider range of complex problems.

Overview

The architecture of MDAgent is illustrated in Fig. 2a, structured around an intelligent agent system, comprising three primary components, namely, the Manager, the Executor, the Planner, the Expert Team (which includes software-specific Workers and Evaluators), and additional supporting elements. MDAgent also includes a memory module, allowing LLMs to remember and track system-wide messages, thus ensuring continuity and context retention throughout complex tasks. In addition, a user interface (UI) was designed to facilitate ongoing interactions between the user and the framework. When the agent is unable to complete a task independently, it can pause the current process and request clarification or assistance from the user via the UI.

Manager

The Manager serves as the entry point, receiving detailed user requirements as an input. When the user's requirements are unclear or lack technical specificity, the Manager refines and rephrases them using expertise from the materials science domain.

Internally, the Manager will invoke the Planner to formulate a high-level plan based on the user's needs. This plan is then broken down into several sub-tasks using expert knowledge, which are subsequently assigned to the agent teams for execution. After every step, the Manager evaluates every team's response and the current plan to determine the next team assigned for execution.

Worker

In the case of LAMMPS, the LammmpsWorker generates LAMMPS script content based on assigned sub-tasks. This process leverages material-specific parameters, such as interatomic potentials, force fields, and simulation protocols, which are essential for accurate MD simulations. Therefore, the Worker should be an LLM with domain knowledge. For the experiments in this paper, fine-tuning techniques were chosen to enhance the domain knowledge of the LLM. The *LSCF-Dataset* was used as the fine-tuning dataset, and the QLoRA technique was applied for fine-tuning, along with the “*Unsloth*”³⁶ framework to improve training efficiency.

Considering that MDAgent may be used for other complex problems in future work, this paper provides RAG and Agent Tools as additional optional features to further ensure the scalability of the project, which also allows LLMs to acquire additional domain-specific knowledge through RAG and Agent Tools. On the one hand, even after fine-tuning the LLM, it may happen that the worker cannot easily solve some problems. At this point the MDAgent is allowed to try to acquire knowledge to help before answering by calling the RAG, or other Tools. On the other hand, considering other problems that the MDAgent faces in the future, the scenario may not be able to find a suitable fine-tuning dataset to fine-tune, which can only be solved by using the RAG and the TOOLS as alternatives.

Simply put, if the Worker needs to reference domain-related content, it can utilize RAG function or other tools to retrieve information from a pre-built knowledge base provided by the user. To demonstrate, for the LammmpsWorker, we gathered a collection of MD papers from the web and the official LAMMPS documentation as resources for the RAG module.

Technically, the RAG system vectorizes the data using an open-source embedding model, then splits the data based on token count. An LLM is subsequently used to automatically annotate keywords, extracting N keywords for each chunk to increase their ranking for queries containing those keywords. During retrieval, the system employs a hybrid approach, integrating both vector retrieval and keyword search to enhance search precision. Furthermore, we utilize an open-source re-ranking model to refine the initial search results, ensuring that the retrieved information is highly relevant to the query. The Tools module includes a *web search tool* and *APIs to access major materials science databases* such as the Materials Project, providing comprehensive datasets on material properties, crystallography, and computational predictions.

Evaluator and executor

MDAgent used a subset of the *LEQS-dataset* designated for fine-tuning as the basis, converting it into an appropriate format for fine-tuning. We then fine-tuned a LammmpsEvaluator model, built on the open-source large models, specifically for evaluation tasks. At the same time, we informed LammmpsEvaluator about the evaluation dimensions and point deduction criteria previously discussed by the experts via PROMPT. To enable MDAgent to autonomously detect and correct code errors, we incorporated principles from Reflexion³⁷ into the interactions between the Worker and Evaluator. The output generated by the Worker is assessed by the Evaluator, which provides a JSON-formatted score and reasoning. If the score falls below a specified threshold, the Evaluator offers feedback and suggestions, guiding the Worker to regenerate the output. This process is repeated until the score exceeds the threshold, at which point the output is considered satisfactory.

For the Executor, we installed the required software in a Docker container, which allows the Manager to interact with and operate within the Docker environment using a command line interface (such as a UNIX

shell) or Python. This allows for the use of command-line commands and the execution of Python code within the environment.

This setup theoretically enables automation by saving the generated script files locally and running them automatically to obtain actual execution results. However, due to certain uncertainties inherent in current LLMs, as well as limitations related to the model size and operational costs, interacting with the Docker container can be resource intensive. In general, it is often more cost-effective for users to independently run the generated scripts in the software.

Data availability

The code and datasets supporting the findings of this study are available from the corresponding author, Xiang Jing (jingxiang@pku.edu.cn), upon reasonable request. The data is also publicly available at https://github.com/FredericVAN/PKU_MDAgent

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Author contributions

Z.F.S. conceived the experiments, wrote the paper, performed specific fine-tuning work, designed and conducted evaluation experiments, and was responsible for the Agent framework design and implementation, as well as assigning tasks to team members. C.X.X. wrote the paper, collected and produced dataset, conducted numerical experiments, graphed the results of the experiments. T.H. performed specific fine-tuning work. Y.T.J. performed specific fine-tuning work. B.W.W. handled material domain tasks in the experiments, evaluated experimental results. X.Y.C. wrote the paper, provided materials science expertise to enhance the clarity and depth of the paper. All of the authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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